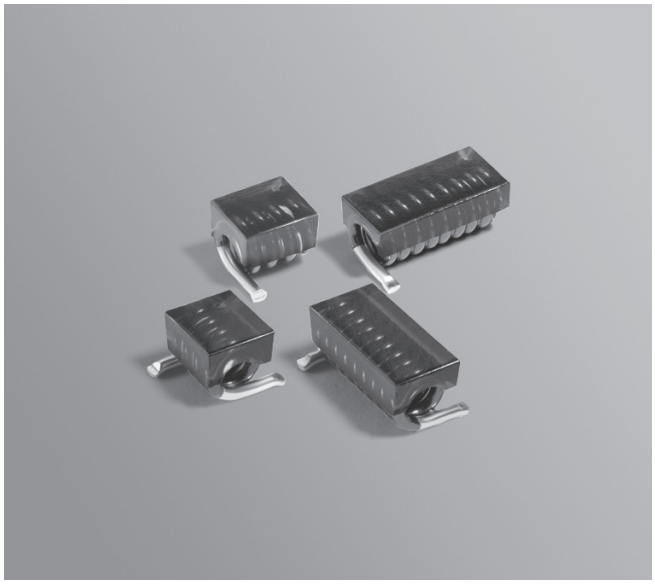
RoHS/ REACH COMPLIANT

Halogen Free

Mini Air Core Inductors 1512SP 2712SP



- Small air core inductors feature high Q and tight tolerances
- Acrylic jacket provides a flat top for pick and place
- Solder coated leads ensure reliable soldering
- Wide range of standard EIA inductance values
- Available in two sizes: 4.45 × 3.94 × 3.15 mm and 6.86 × 4.45 × 3.15 mm
- **Designer's Kit C501** contains 10 of each part

Terminations RoHS compliant tin-silver over copper. Other terminations available at additional cost.

Ambient temperature –40°C to +125°C with Irms current, +125°C to +140°C with derated current

Storage temperature Component: –40°C to +140°C. Tape and reel packaging: –40°C to +80°C

Resistance to soldering heat Max three 40 second reflows at +260°C, parts cooled to room temperature between cycles

Temperature Coefficient of Inductance (TCL) +5 to +70 ppm/°C

Moisture Sensitivity Level (MSL) 1 (unlimited floor life at <30°C / 85% relative humidity)

PCB washing Tested to MIL-STD-202 Method 215 plus an additional aqueous wash. See Doc787_PCB_Washing.pdf.

Part number[1]	L[2] (nH)	Percent tol[3]	Q[4] min	SRF typ[5] (GHz)	DCR max[6] (mOhm)	Irms[7] max (A)	Weight (mg)
1512SP-2N5K1E_	2.5	**10**	80	14.0	1.0	11.1	37.8
1512SP-2N7K1E_	2.7	**10**	80	14.0	3.0	8.4	16.6
1512SP-4N7_2E_	4.7	**5,2**,1	80	18.0	2.0	9.4	51.3
1512SP-5N6_2E_	5.6	**5,2**,1	80	18.0	3.2	6.6	33.9
1512SP-6N8_3E_	6.8	**5,2**,1	80	6.4	2.1	9.8	78.4
1512SP-8N2_3E_	8.2	**5,2**,1	80	7.2	2.9	7.4	57.2
1512SP-10N_4E_	10.0	**5,2**,1	80	3.4	2.5	9.2	94.8
1512SP-12N_4E_	12.0	**5,2**,1	80	4.1	3.5	8.0	80.2
1512SP-15N_4E_	15.0	**5,2**,1	80	4.5	5.0	5.9	55.8
1512SP-18N_4E_	18.0	**5,2**,1	80	5.2	9.3	3.7	34.2
1512SP-22N_5E_	22.0	**5,2**,1	80	3.1	5.3	5.2	64.0
1512SP-27N_5E_	27.0	**5,2**,1	80	4.2	10.6	3.5	37.5
1512SP-33N_6E_	33.0	**5,2**,1	80	2.8	8.6	3.9	50.7
1512SP-43N_7E_	43.0	**5,2**,1	80	2.1	10.6	4.3	55.5
1512SP-51N_7E_	51.0	**5,2**,1	80	2.1	14.5	2.7	48.0
1512SP-62N_8E_	62.0	**5,2**,1	80	1.8	15.5	2.7	51.1

Continued on next page

1. When ordering, please specify **tolerance**, **termination**, and **packaging** codes:

1512SP-62NJ8EC

Tolerance: **F** = 1% **G** = 2% **J** = 5% **K** = 10% (Table shows stock tolerances in bold.)
Termination: **E** = Halogen free component. RoHS compliant tin-silver (96.5/3.5) over copper terminations. Special order: **T** = RoHS tin-silver-copper (95.5/4/0.5) or **S** = non-RoHS tin-lead (63/37).
Packaging: **C** = 7″ machine-ready reel. EIA-481 embossed plastic tape, 700 (1512SP) or 500 (2712SP) parts per full reel. Quantities less than full reel available: in tape (not machine ready) or with leader and trailer ($25 charge).
D = 13″ machine-ready reel. EIA-481 embossed plastic tape. Factory order only, not stocked, 2500 (1512SP) or 2200 (2712SP) parts per full reel.

2. Inductance measured at 150 MHz with an Agilent/HP 4286 or equivalent, a Coilcraft SMD-A and correlation.
3. Tolerances in bold are stocked for immediate shipment.
4. Q measured at 150 MHz using an Agilent/HP 4291A with an Agilent/HP 16193A test fixture or equivalents.
5. SRF is measured on an Agilent 8753ES or equivalent with a Coilcraft CCF1268 test fixture.
6. DCR is measured on a Keithley 580 Micro-ohmmeter or equivalent.
7. Current that causes a 15°C temperature rise from 125°C ambient.
8. Electrical specifications at 25°C.

Refer to Doc 362 "Soldering Surface Mount Components" before soldering.

Coilcraft®
www.coilcraft.com

US +1-847-639-6400 sales@coilcraft.com
UK +44-1236-730595 sales@coilcraft-europe.com
Taiwan +886-2-2264 3646 sales@coilcraft.com.tw
China +86-21-6218 8074 sales@coilcraft.com.cn
Singapore + 65-6484 8412 sales@coilcraft.com.sg

© Coilcraft Inc. 2024
This product may not be used in medical or high risk applications without prior Coilcraft approval
Specification subject to change without notice
Please check web site for latest information

Mini Air Core Inductors – 2712SP and 1512SP

Part number[1]	Inductance[2] (nH)	Percent tolerance[3]	Q[4] min	SRF typ[5] (GHz)	DCR max[6] (mOhm)	Irms[7] max (A)	Weight (mg)
2712SP-10N_4E_	10.0	**5,2**,1	85	6.6	3.1	10.3	89.5
2712SP-12N_5E_	12.0	**5,2**,1	85	5.8	3.4	7.8	102.5
2712SP-15N_6E_	15.0	**5,2**,1	85	4.9	3.3	8.2	142.6
2712SP-18N_7E_	18.0	**5,2**,1	85	3.9	3.5	7.9	156.2
2712SP-22N_7E_	22.0	**5,2**,1	85	3.9	4.5	7.0	132.4
2712SP-27N_8E_	27.0	**5,2**,1	85	3.1	5.0	6.6	144.0
2712SP-33N_8E_	33.0	**5,2**,1	85	3.3	7.4	5.1	98.5
2712SP-39N_9E_	39.0	**5,2**,1	90	2.7	6.8	4.6	133.3
2712SP-47N_0E_	47.0	**5,2**,1	90	2.4	8.0	4.8	141.6
2712SP-51N_0E_	51.0	**5,2**,1	95	2.4	9.25	4.1	117.9
2712SP-56N_0E_	56.0	**5,2**,1	95	2.6	15.5	2.9	79.3
2712SP-68N_1E_	68.0	**5,2**,1	100	2.3	16.0	3.1	85.9
2712SP-82N_3E_	82.0	**5,2**,1	100	1.9	14.7	2.8	118.3
2712SP-101_4E_	100.0	**5,2**,1	100	1.8	19.0	2.3	103.2
2712SP-121_4E_	120.0	**5,2**,1	100	1.6	26.5	2.3	87.9
2712SP-151_6E_	150.0	**5,2**,1	100	1.4	29.0	2.2	95.6

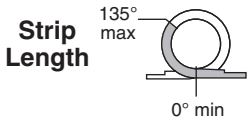
1. When ordering, please specify **tolerance**, **termination**, and **packaging** codes:

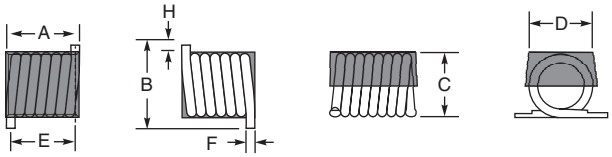
2712SP-151J6EC

Tolerance: **F** = 1% **G** = 2% **J** = 5% **K** = 10% (Table shows stock tolerances in bold.)
Termination: **E** = Halogen free component. RoHS compliant tin-silver (96.5/3.5) over copper terminations.
Special order: **T** = RoHS tin-silver-copper (95.5/4/0.5) or **S** = non-RoHS tin-lead (63/37).
Packaging: **C** = 7″ machine-ready reel. EIA-481 embossed plastic tape, 700 (1512SP) or 500 (2712)SP parts per full reel. Quantities less than full reel available: in tape (not machine ready) or with leader and trailer ($25 charge).
D = 13″ machine-ready reel. EIA-481 embossed plastic tape. Factory order only, not stocked, 2500 (1512SP) or 2200 (2712SP) parts per full reel.

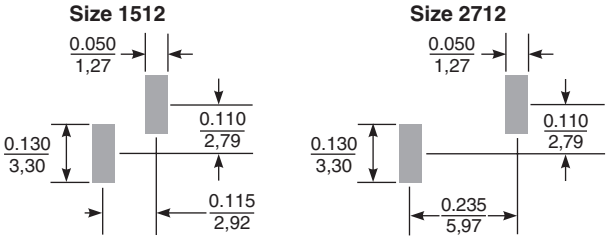
2. Inductance measured at 150 MHz with an Agilent/HP 4286 or equivalent, a Coilcraft SMD-A and correlation.
3. Tolerances in bold are stocked for immediate shipment.
4. Q measured at 150 MHz using an Agilent/HP 4291A with an Agilent/HP 16193A test fixture or equivalents.
5. SRF is measured on an Agilent 8753ES or equivalent with a Coilcraft CCF1268 test fixture.
6. DCR is measured on a Keithley 580 Micro-ohmmeter or equivalent.
7. Current that causes a 15°C temperature rise from 125°C ambient.
8. Electrical specifications at 25°C.

Refer to Doc 362 "Soldering Surface Mount Components" before soldering.





Recommended Land Patterns



Dimensions are in inches / mm

Part Size	Units	A MAX	B MAX	C MAX	D	E	F MAX	H
1512	in	0.155	0.175	0.124 0.130 (-2N5)	0.110 ±0.010	0.115 ±0.010	0.030	0.020 ±0.010
	mm	3.94	4.45	3.15 3.30 (-2N5)	2.79 ±0.25	2.92 ±0.25	0.76	0.51 ±0.25
2712	in	0.270	0.175	0.124	0.110 ±0.010	0.230 ±0.015	0.030	0.020 ±0.010
	mm	6.86	4.45	3.15	2.79 ±0.25	5.84 ±0.38	0.76	0.51 ±0.25

Packaging
1512: 700/7″ reel; 2500/13″ reel Plastic tape: 12 mm wide, 0.32 mm thick, 8 mm pocket spacing, 3.3 mm pocket depth
2712: 500/7″ reel; 2200/13″ reel Plastic tape: 16 mm wide, 0.28 mm thick, 8 mm pocket spacing, 3.4 mm pocket depth
Recommended pick and place nozzle: OD: 0.110″; ID: 0.091″

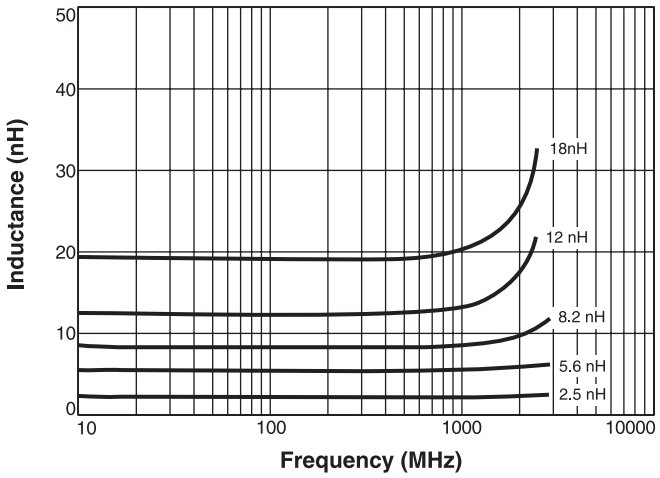
Coilcraft®
www.coilcraft.com

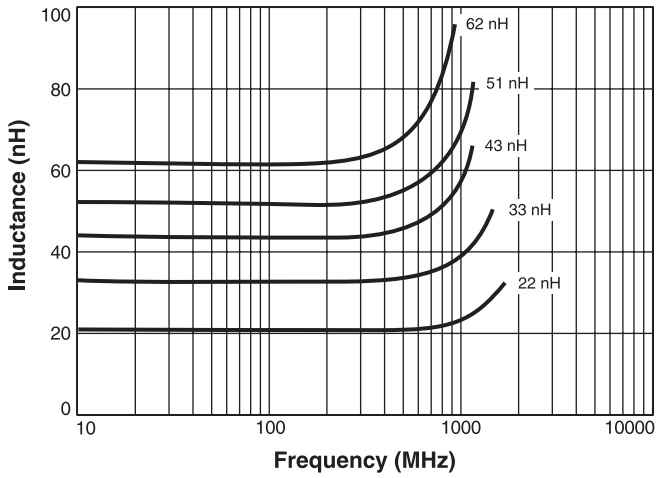
US +1-847-639-6400 sales@coilcraft.com
UK +44-1236-730595 sales@coilcraft-europe.com
Taiwan +886-2-2264 3646 sales@coilcraft.com.tw
China +86-21-6218 8074 sales@coilcraft.com.cn
Singapore + 65-6484 8412 sales@coilcraft.com.sg

© Coilcraft Inc. 2024
This product may not be used in medical or high risk applications without prior Coilcraft approval
Specification subject to change without notice
Please check web site for latest information

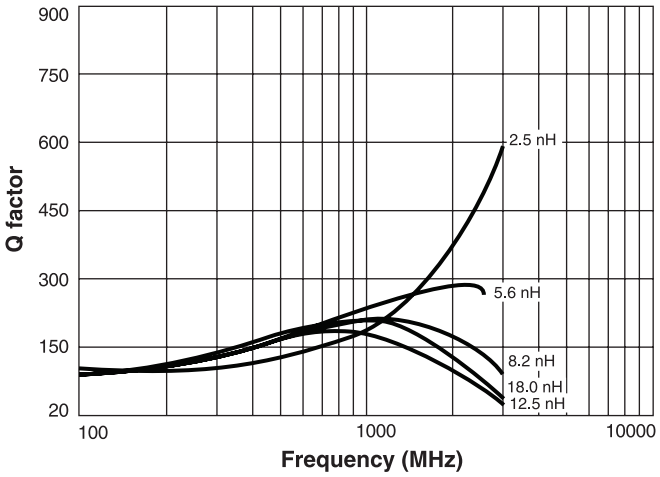
Mini Air Core Inductors – 1512SP

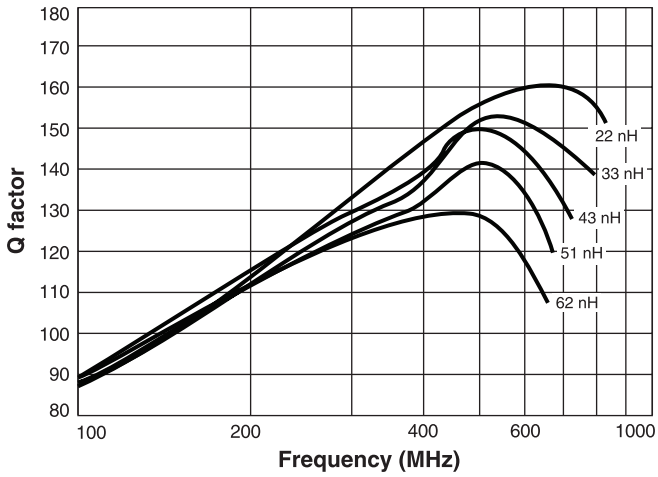
Typical L vs Frequency





Typical Q vs Frequency





www.coilcraft.com

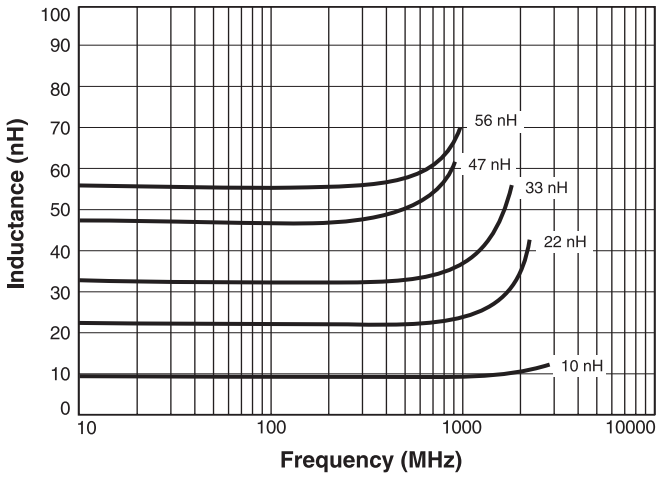
US +1-847-639-6400 sales@coilcraft.com
UK +44-1236-730595 sales@coilcraft-europe.com
Taiwan +886-2-2264 3646 sales@coilcraft.com.tw
China +86-21-6218 8074 sales@coilcraft.com.cn
Singapore + 65-6484 8412 sales@coilcraft.com.sg

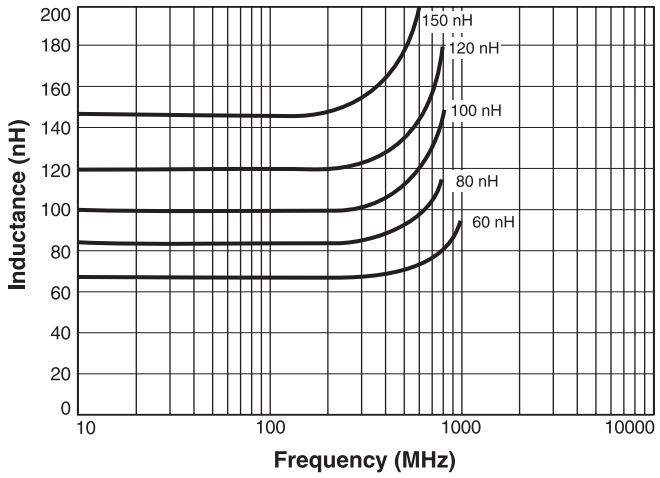
Document 1536-3 Revised 08/27/24

© Coilcraft Inc. 2024
This product may not be used in medical or high risk applications without prior Coilcraft approval
Specification subject to change without notice
Please check web site for latest information

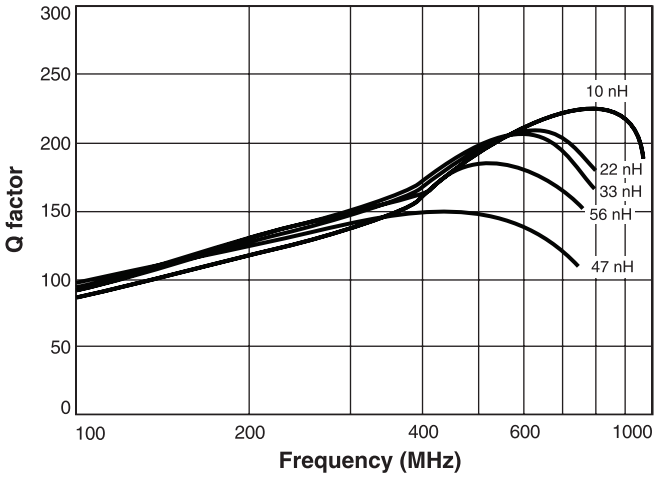
Mini Air Core Inductors – 2712SP

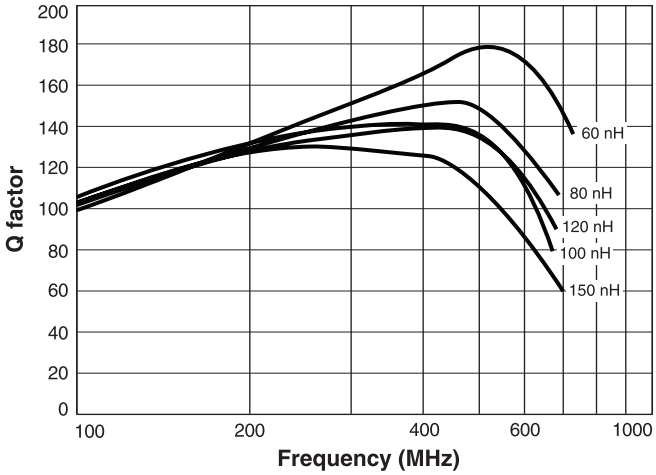
Typical L vs Frequency





Typical Q vs Frequency





www.coilcraft.com

US +1-847-639-6400 sales@coilcraft.com
UK +44-1236-730595 sales@coilcraft-europe.com
Taiwan +886-2-2264 3646 sales@coilcraft.com.tw
China +86-21-6218 8074 sales@coilcraft.com.cn
Singapore + 65-6484 8412 sales@coilcraft.com.sg

Document 1536-4 Revised 08/27/24

© Coilcraft Inc. 2024
This product may not be used in medical or high risk applications without prior Coilcraft approval
Specification subject to change without notice
Please check web site for latest information